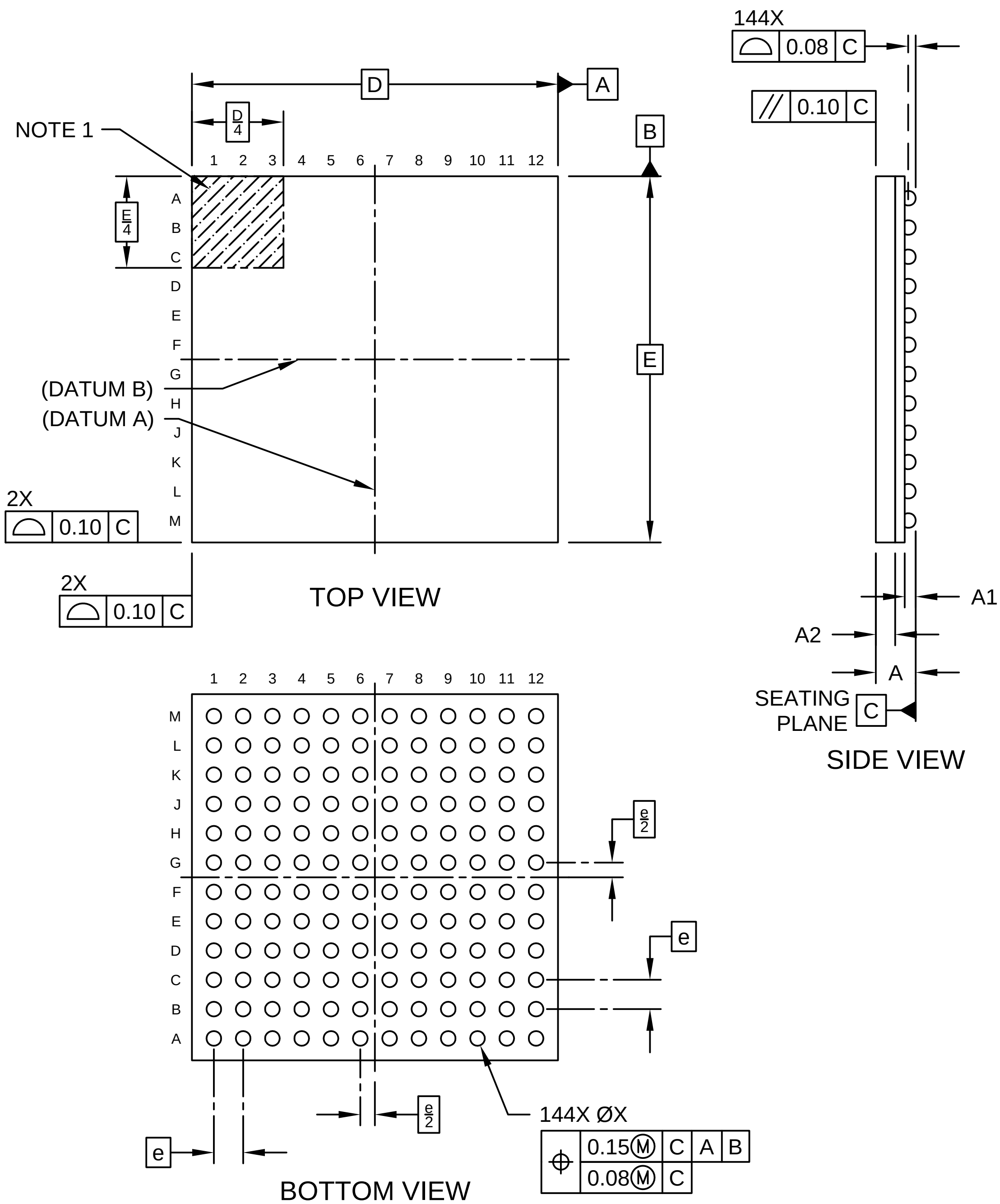
144-Ball Thin Fine Pitch Ball Grid Array (7LX) - 10x10 mm Body [TFBGA]

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-490 Rev A Sheet 1 of 2

144-Ball Thin Fine Pitch Ball Grid Array (7LX) - 10x10 mm Body [TFBGA]

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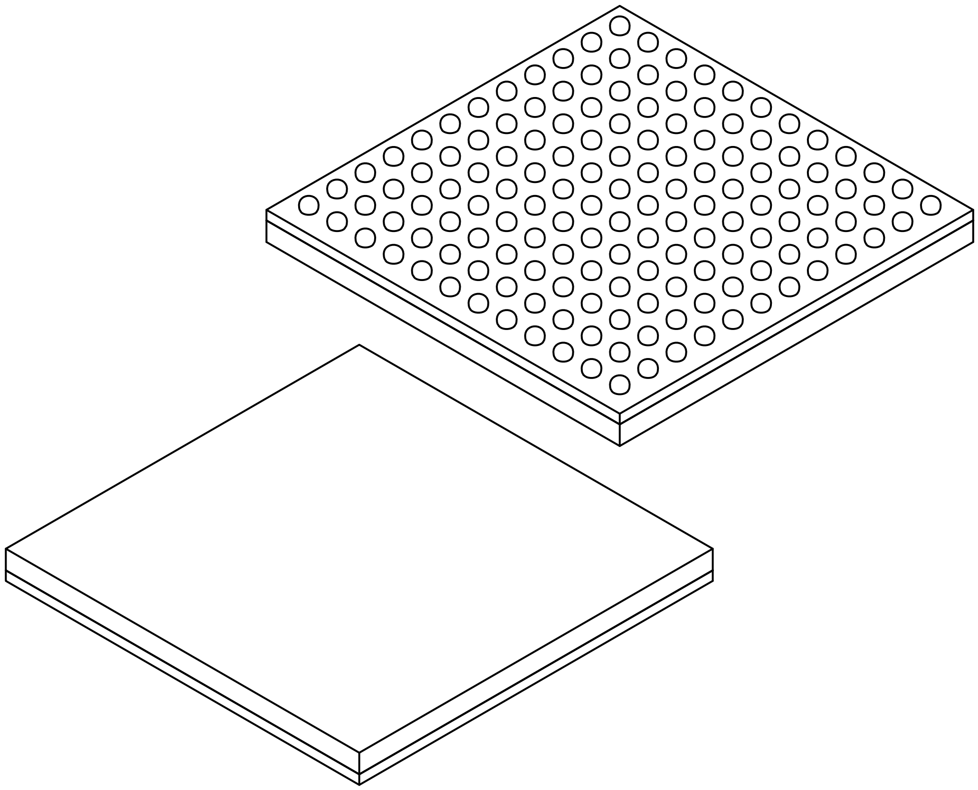


Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		144	
Pitch	e		0.80 BSC	
Overall Height	A	-	-	1.19
Ball Height	A1	0.21	0.30	-
Mold Cap Thickness	A2	0.48	0.53	0.58
Overall Length	D		10.00 BSC	
Overall Width	E		10.00 BSC	
Ball Diameter	b	0.35	0.40	0.45

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M

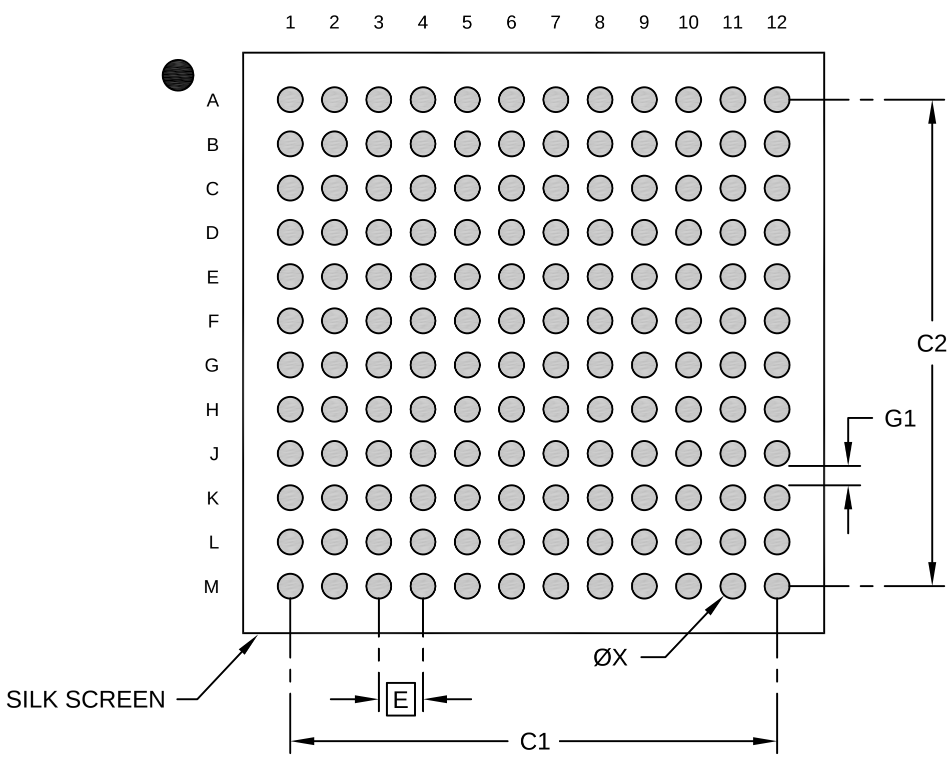
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-490 Rev A Sheet 2 of 2

144-Ball Thin Fine Pitch Ball Grid Array (7LX) - 10x10 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		8.80	
Contact Pad Spacing	C2		8.80	
Contact Pad Diameter (X144)	X			0.45
Contact Pad to Contact Pad	G1	0.35		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2490 Rev A